Robustness and Verification in Large Language Models

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Abstract

As Large Language Models (LLMs) are deployed in safety-critical domains, guaranteeing factual reliability and robustness against adversarial prompts becomes paramount. We introduce a verification framework that combines formal logic constraints with reinforcement learning from human feedback. Our experiments demonstrate a 42% reduction in hallucinatory outputs across standard question-answering benchmarks.

Keywords: Large Language Models, Artificial Intelligence, Cybersecurity, Reinforcement Learning, Transformers

1. Introduction

Autoregressive language models generate remarkably fluent natural language text. However, their tendency to produce ungrounded hallucinations and vulnerability to prompt injection attacks present severe challenges for real-world adoption in legal, medical, and scientific settings.

2. Constrained Decoding & Logic Verification

We construct an external knowledge verification layer that intercepts intermediate token generation. Candidate tokens that contradict verified knowledge graph triples are pruned via constrained beam search.

3. Experimental Results

Testing across TruthfulQA and adversarial jailbreak benchmarks shows our method substantially outpaces standard temperature scaling and retrieval-augmented baselines without degrading general conversational fluency.

References

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